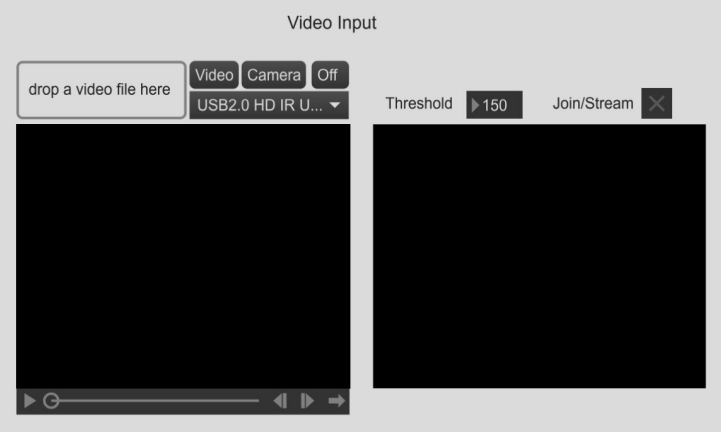
Documentation – instrument-1

Controls

Video input



drop a video file here – Text in the field where any video can be dragged and dropped

Video – Switches the tool's video input to video (must be inserted first)

Camera – Switches the tool's video input to camera (requires selecting a camera via the **Dropdown Menu**)

Off – Turns off the video input

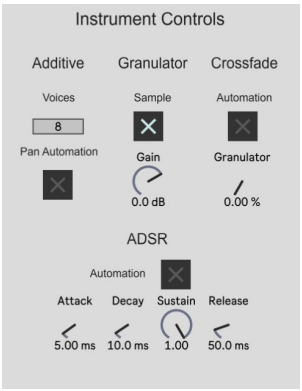
Dropdown Menu – Allows selection of a video source for the **Camera** option

Playbar – Control element for playing, stopping, and looping the inserted video (functional only for **Video** input)

Threshold – Sets the boundary between black and white pixels

Join/Stream – Choose between two methods of sending data to the tool's parameters

Instrument controls



Voices (Additive) – Sets the maximum number of voices (notes that can be played simultaneously)

Pan Automation (Additive) – Enables/disables spatial distribution control of individual harmonic components from video data

Sample (Granulator) – Enables/disables sampling of the output signal from the additive synthesis source

Gain (Granulator) – Adjusts the gain of the granulator signal

Automation (Crossfade) – Enables/disables automation of signal blending between the additive synthesizer and granulator using video data

Granulator (Crossfade) – Determines how loud the granulator will be relative to the additive synthesizer

Automation (ADSR) – Enables/disables amplitude envelope automation for the additive synthesizer

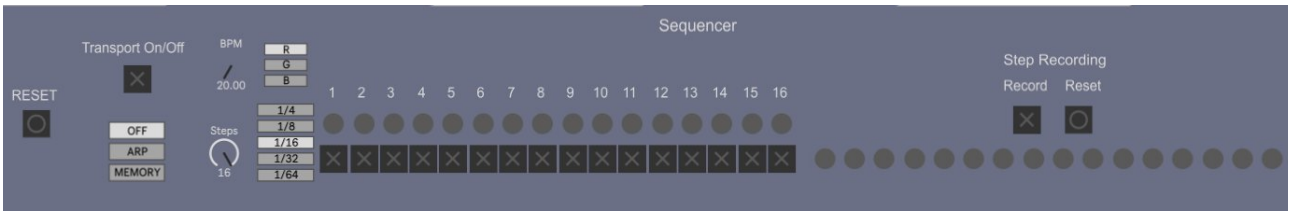
Attack (ADSR) – **Attack** parameter setting

Decay (ADSR) – **Decay** parameter setting

Sustain (ADSR) – **Sustain** parameter setting

Release (ADSR) – **Release** parameter setting

Sequencer



Reset – Returns the step to position 1

Transport On/Off – Turns the Transport (and sequencer) on/off

BPM – Displays the current tempo

Steps – Sets the current number of sequencer steps

R, G, B switch – Selects the layer for obtaining data from the video

1/4, 1/8, 1/16, 1/32, 1/64 switch – Sets the duration of one step in the sequencer

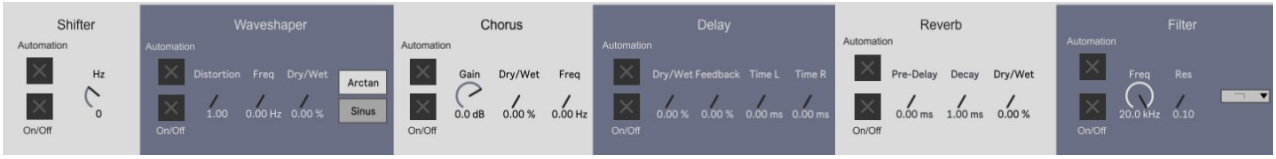
OFF, ARP, MEMORY switch – Switches between the sequencer being off, ARP mode (while holding a note, generates pitch for individual enabled steps of the sequence using video data), and MEMORY mode (allows recording a sequence that is then played back)

On/Off switch (for individual steps) – Turns a given step on/off

Record (Step Recording) – Turns step sequence recording on/off

Reset (Step Recording) – Resets step recording to position 1

Effects unit



Automation (all effects) – Turns on/off parameter control using video data

On/Off (all effects) – Turns the effect on/off

Dry/Wet (all effects with an exception of Frequency shifter and Filter) – Sets the ratio between processed and unprocessed signal

Hz (Frequency shifter) – Determines how many Hz the spectrum is shifted (range: -1500 Hz to 3000 Hz)

Distortion (Waveshaper) – Sets the value by which the signal is multiplied in a nonlinear function (range: 1 to 30)

Freq (Waveshaper) – Sets the frequency of a band-stop filter (range: 0 Hz to 20 kHz)

Arctan, Sinus switch (Waveshaper) – Determines the function the processed signal passes through

Gain (Chorus) – Adjusts the output volume of the effect (range: -∞ dB to +30 dB)

Freq (Chorus) – Sets the frequency of the low-frequency oscillator (range: 0.1 Hz to 20 Hz)

Feedback (Delay) – Sets the feedback gain (range: 0 to 1)

Time L, Time R (Delay) – Sets the delay for the right and left channels (range: 0 ms to 10 s)

Pre – Delay (Reverb) – Sets the delay before the first reflection arrives (range: 0 ms to 1 s)

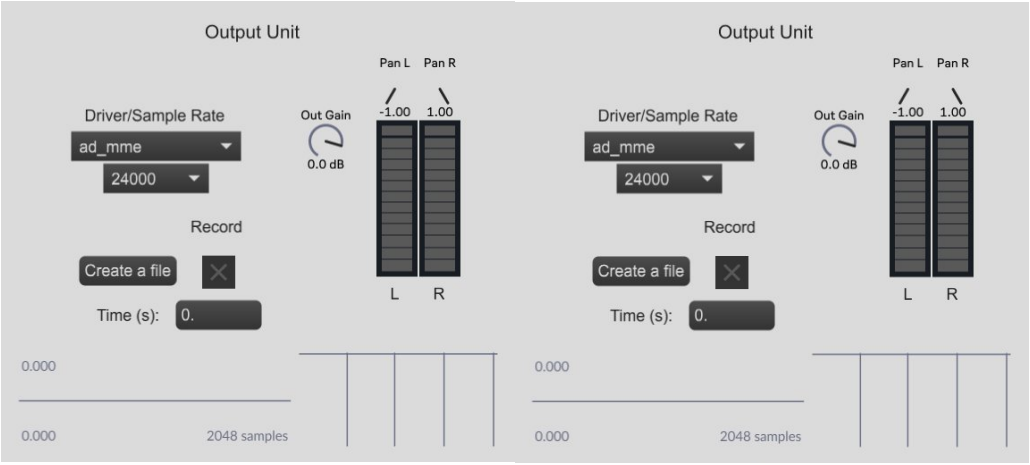
Decay (Reverb) – Sets the Decay parameter of the Reverb effect (range: 1 ms to 60 s)

Freq (Filter) – Sets the filter frequency (range: 0 Hz to 20 kHz)

Res (Filter) – Sets the filter resonance (cannot be influenced by video parameters)

Type (Filter) – Sets the filter type (options: low-pass, high-pass, band-pass, and band-stop; cannot be influenced by video data)

Output unit



Out Gain – Potentiometer for adjusting the output signal volume

Pan L a Pan R – Potentiometers for setting the distribution of the left and right channel signals in the stereo base

Create a file – Clicking opens a menu to create and save a recorded audio file

Record – Clicking starts recording (a file must first be created using the **Create a file** control), clicking off stops recording

Driver/Sample Rate – Use the dropdown menu to select an audio driver and sample rate

Initial setup

To start the instrument, you need a **connected MIDI controller** that sends Note On and Note Off messages. Additionally, you need **to turn on any available video source** (see the labeled controls).

